Convergence of Observer Dynamics in the Ruliad: From Conjecture to Theorem

Sam A. Senchal [*]

Open Research Institute

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Abstract

The God Conjecture (Senchal, 2025b) claims that all observers in a computational universe converge toward persistent structures through iterated entropy reduction as a consequence of sampling. This claim was stated as a conjecture with four proof sketches. This paper provides the formal proof. The architecture proceeds in three stages: (i) a finite-time convergence theorem for deterministic observers, showing that every orbit of the entropy reduction functor terminates at a fixed point within $|R_O|$ steps; (ii) an exponential convergence theorem for stochastic observers via the Dobrushin contraction coefficient, extending to non-stationary environments under weak ergodicity; and (iii) a structural observation noting that all observer fixed points admit unique morphisms to the terminal object TI by its terminality, with the full categorical construction deferred to a companion paper. We prove the MetaObserver property from the Ruliad's definition rather than assuming it. The substantive mathematical content resides in the local convergence theorems (i–ii); the structural observation (iii) provides the interpretive frame connecting these results to the God Conjecture's claims. The result holds under one substantive physical assumption—that the universe has stable statistical properties—with all other conditions either physically justified or definitional.

Contribution Statement. The proof techniques are standard (finite descent, Dobrushin contraction, Banach fixed-point). The contribution is the formalisation itself: demonstrating that observer dynamics, as defined in Senchal (2025a), satisfy the conditions under which these techniques apply, and the structural observation connecting local convergence to global structure via TI's terminality.

Keywords: Observer Theory, Ruliad, convergence, fixed point, Dobrushin coefficient, terminal object, Free Energy Principle, entropy reduction

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1 Introduction

1.1 The Gap

Observer Theory and the Ruliad: An Extension to the Wolfram Model (Senchal, 2025a) formalises observers as sampling functors on the Ruliad—the entangled limit of all possible computations (Gorard and Arsiwalla, 2021; Wolfram, 2021). The theory establishes that observers reduce entropy through coarse-graining (via the Data Processing Inequality), that this process creates persistent structures (stable patterns invariant under further observation), and that the discovery of persistent structures expands the observer's accessible field.

What the theory does not prove is the capstone claim: that this process converges. That observers settle to an equilibrium. That the equilibrium is unique under noise. And that all observers—regardless of architecture, constraint profile, or starting conditions—converge toward structurally connected persistent structures.

The God Conjecture (Senchal, 2025b) advanced this claim with four proof sketches. The paper was explicit about the gap: "The proof is beyond the scope of this paper, but it relies on the finiteness or compactness of state space and the Observer's limits" (Senchal, 2025a, Appendix B.3).

This paper closes the gap. The local convergence theorems (Sections 5–6) contain the substantive mathematical content. The structural observation (Section 8) provides the interpretive frame connecting these results to the God Conjecture's broader claims.

1.2 Why Convergence Matters

If the convergence theorem holds, several consequences follow that do not follow from Observer Theory alone:

First, the persistent structures that observers discover (physical laws, mathematical truths, logical relations) are not contingent features of human cognition. They are attractors of the dynamics. Any finite observer in any computational universe with stable statistics will discover them, because they are fixed points of the entropy reduction process. Different observer architectures will converge on the same structures—not because they share cognitive architecture, but because the structures are features of the territory, not the map.

Second, the convergent features of religious and ethical traditions identified in the God Conjecture—the independent arrival at unified infinite grounds (Ein Sof, Brahman, Dao, Logos)—become formally predicted rather than merely observed. Whether the terminal categorical limit TI corresponds to what these traditions describe is an interpretive question the mathematics makes precise but does not settle (Section 9.6).

Third, the ratchet effect—the claim that the inventory of persistent structures is monotonically non-decreasing at the observer-class level—graduates to theorem.

Fourth, the theorem connects the Observer Theory Extension to the Free Energy Principle (FEP) at the level of convergence results. FEP proves convergence to a non-equilibrium steady state via a Lyapunov function (Friston, 2019). Theorem 5.2 proves convergence via an independent route.

1.3 Structure of the Proof

The convergence claim separates into two logically distinct components:

The mathematical content (Theorem 5.2–6.11): Each observer's iterated entropy reduction converges to a distribution supported on persistent structures. These are genuine dynamical results using the Data Processing Inequality, finiteness, and Markov chain theory.

The structural observation (Observation 8.1, Theorem 8.4(ii)): All observer equilibria are structurally connected to TI. The existence of unique morphisms from fixed points to TI is a structural consequence of TI's terminality (i.e. not an independent dynamical theorem). The convergence theorems tell us that observers *reach* persistent structures; TI's defining property tells us that persistent structures *point toward* TI. These are logically independent facts with independent sources. The full categorical treatment—constructing TI from observer dynamics via a filtered colimit—is deferred to a companion paper (Senchal, 2026b).

Convergence comes from observation. Uniqueness comes from reality. More precisely: convergence holds for any observer (it depends only on finiteness); structural connection to TI holds for any connected computational substrate (it depends only on the Ruliad's topology). Together: all finite observers in a connected universe converge on persistent structures that are structurally related to the same categorical limit.

The paper treats two settings. For deterministic observers, convergence is finite-time ($\leq |R_O|$ steps). For stochastic observers, convergence is exponential with a unique limit via the Dobrushin contraction coefficient. The deterministic result says convergence happens; the stochastic result says how fast and to what.

2 Preliminaries

Each definition answers a question raised by the previous one. Table 1 provides a summary of all key terms.

Definition 2.1 (Ruliad)**.** The Ruliad $\mathbf{R}$ is a category whose objects are computational states and whose morphisms are computational transitions generated by all possible rules, forming an ∞-groupoid in the limit (Gorard and Arsiwalla, 2021).

The Ruliad is the space of everything that could be computed. It is infinite and overwhelmingly inaccessible to any finite entity. Think of it as a library containing every possible book, every possible computation, every possible physical law—but most of the library is unreachable from any given starting point. This creates the need for an Observer.

Definition 2.2 (Observer)**.** An Observer O is defined by a sampling functor $S_O \colon \mathbf{R} \to R_O$, constrained by boundedness B_O, persistence P_O, and relevance Rel_O.

An observer is any entity—biological, digital, or otherwise—that samples a finite portion of the Ruliad. The sampling functor is the mathematical name for the observer's "window" onto reality: it maps the full computational universe to the subset the observer can actually perceive and process.

Definition 2.3 (Entropy Reduction Functor)**.** $ER \colon R_O \to R_O$ sends each state to its entropy-reduced form. A state x is *persistent* when $H_{\text{red}}(x) = H(x)$, denoted $x \in \mathrm{Fix}(ER)$.

Observation compresses. Each time an observer samples a state, it coarse-grains—throws away microstates it cannot distinguish—and thereby reduces entropy. A persistent structure is a state that observation cannot compress further: it is already maximally reduced. Mathematical truths, physical constants, and logical tautologies are examples.

Table 1: Key terms and definitions.

Term	Symbol	Definition	Status
Ruliad	$\mathbf{R}$	Category of all computational states and transitions; ∞-groupoid in the limit	Definitional
Observer	O	Sampling functor $S_O\colon \mathbf{R} \to R_O$, constrained by $(B_O, P_O, \mathrm{Rel}_O)$	Definitional
Observable Ruliad	R_O	Image of S_O; the finite set of states accessible to O	Derived
Entropy reduction	ER	$ER\colon R_O \to R_O$, sends states to their entropy-reduced form	Definitional
Persistent structure	$x \in \mathrm{Fix}(ER)$	State invariant under ER: $H(ER(x)) = H(x)$	Derived
Terminal object	TI	$\forall X,\ \exists!$ morphism $X \to \mathrm{TI}$	Definitional
Dobrushin coeff.	$\delta(K)$	$1 - \min_{x_1,x_2} \sum_y \min(K_{y,x_1}, K_{y,x_2})$	Standard
Noise floor	$\varepsilon_{\min}$	$\min_{x,y}(K_t^{N_{\max}})_{y,x} \geq \varepsilon_{\min}$	Assumed in (C2′c)
NESS	—	Non-Equilibrium Steady State	FEP terminology

Definition 2.4 (Terminal Object)**.** TI is the terminal object in the category of Ruliad states: $\forall\, X \in \mathrm{ob}(\mathbf{T}),\ \exists!\, f\colon X \to \mathrm{TI}$.

A terminal object is one to which every other object has exactly one arrow (morphism). This is a structural property guaranteeing all objects are connected to TI. *Whether observer dynamics follow this connection is a separate question addressed by the convergence theorems. The morphism is categorical; the convergence is dynamical.*

Proposition 2.5 (Entropy Reduction; Senchal 2025a, App. B.2)**.** *For any $x \in R_O$:*

$$H(ER(x)) \ \leq\ H(x), \tag{1}$$

with equality if and only if ER acts as the identity on x.

Coarse-graining cannot increase information—this is a direct consequence of the Data Processing Inequality (Cover and Thomas, 2006, Thm. 2.8.1). The inequality says observation can only reduce or maintain entropy, never increase it.

3 Conditions

Condition 3.1 (C1—Strict Entropy Reduction)**.** For any non-fixed state $x \in R_O \setminus Fix(ER)$:

$$H(ER(x)) \ <\ H(x). \tag{2}$$

Physical status: physically justified, but not automatic. Proposition 2.5 gives $\leq$. Strict inequality requires that ER genuinely compresses—maps multiple states to the same output—rather than merely permuting them. Entropy-preserving transformations (rotations, relabellings) are possible in principle but are transient: a computationally bounded observer

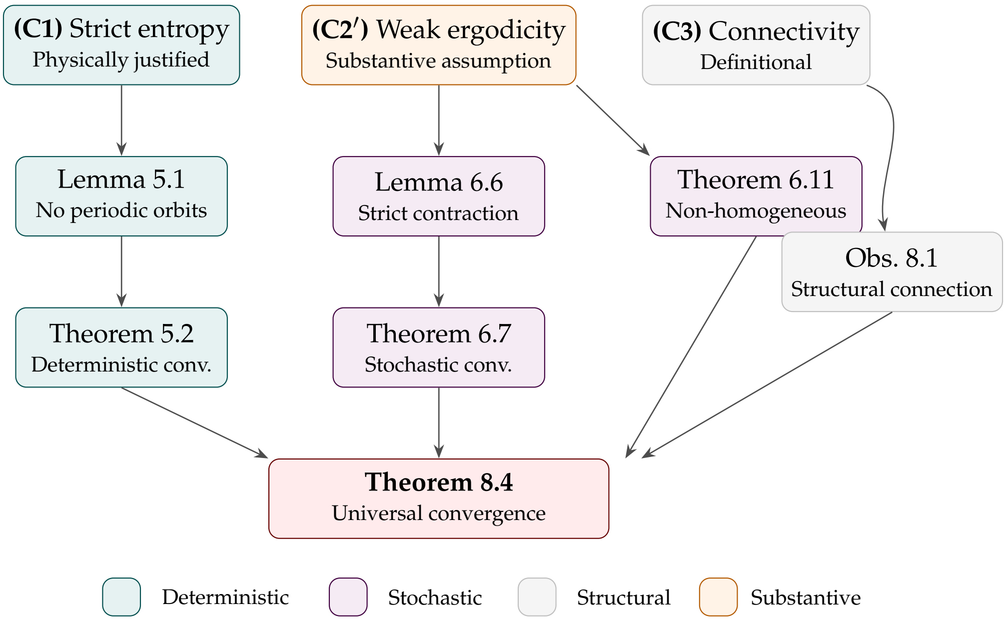

Figure 1: Proof dependency diagram. The three conditions (top row) feed into intermediate lemmas (middle) and the main theorems (bottom), culminating in Theorem 8.4. Colour encodes the proof track: teal for deterministic, purple for stochastic, grey for structural, and amber for the single substantive physical assumption (C2′). The structural track (Obs. 8.1) follows from TI's terminality (Definition 2.4); the full categorical construction is deferred to Senchal (2026b).

($B_O < \infty$) operating on a computationally irreducible environment cannot maintain them indefinitely.

The argument is as follows. Computational irreducibility (Wolfram, 2002, Ch. 12) guarantees that the Ruliad generates novel structure faster than any finite observer can process it. An observer that merely permutes its states without compressing is one whose sampling functor preserves information perfectly—but this requires B_O to match the local complexity of the source, which computational irreducibility makes impossible for all but trivial subsets. Continued sampling forces at least occasional genuine compression. (C1) therefore holds generically: every orbit eventually encounters a strict decrease, which on a finite set suffices for termination.

Analogy: imagine trying to losslessly copy an endlessly growing, infinitely detailed painting using a finite canvas. Eventually, you must simplify—you cannot merely rearrange details forever. That forced simplification is what strict entropy reduction captures.

Condition 3.2 (**C2′**—Weak Ergodicity)**.**

(a) B_O is uniformly bounded.

(b) (C1) holds at every step.

(c) For the stochastic setting: $\exists$ uniform bound $N_{\max}$ on primitivity indices and uniform noise floor $\varepsilon_{\min} > 0$ such that $\min_{x,y}(K_t^{N_{\max}})_{y,x} \geq \varepsilon_{\min}$ for all t.

Physical status: the only substantive assumption. Part (c) requires a common primitive support graph shared by all K_t: the noise floor must connect the same state pairs persistently, justified if the noise source is state-independent (quantum/thermal). Under this condition, Wielandt's bound gives $N_{\max} \leq (n_{\max} - 1)^2 + 1$ per matrix (Levin et al., 2009, Ch. 1). The noise floor $\varepsilon_{\min}$ operates at different scales: significant at the measurement level, orders of magnitude smaller at the macroscopic epistemic level. The convergence theorem holds for any $\varepsilon_{\min} > 0$, but the timescale scales as $O(1/\varepsilon_{\min})$—cross-basin equilibration may require cosmological timescales. Paradigm shifts are rare precisely because the effective macroscopic noise floor is small.

Why this assumption is physically reasonable: quantum mechanics guarantees a non-zero noise floor in any physical measurement process (Zurek, 2003). Thermal fluctuations provide a second, independent source. The assumption fails only if there exist perfectly noiseless observers—entities that can sample without any measurement uncertainty. No known physical system achieves this.

Why this assumption is non-trivial: uniformity is the strong part. It requires that the noise floor does not collapse to zero as the environment changes—that quantum/thermal noise persists across all epochs and all observer states. A universe with regimes of perfect determinism would violate (C2′c). We assume this does not occur, and flag it as the paper's central physical commitment.

Condition 3.3 (**C3**—Ruliad Connectivity)**.** **R** is connected.

Status: definitional (from Definition 2.1).

The Ruliad is connected by construction: every computational state is reachable from every other via some sequence of rule applications. This is not a physical assumption but a consequence of the Ruliad containing all possible rules.

4 Running Examples

Two toy systems illustrate the dynamics before the general theorems. These are referenced throughout the paper.

4.1 System α: Single Fixed Point

$R_O = \{A, B, C, D, E\}$, with ER defined as:

$$A \mapsto A, \quad B \mapsto A, \quad C \mapsto A, \quad D \mapsto C, \quad E \mapsto D.$$

Unique persistent structure: A. Chains: $E \to D \to C \to A$ (3 steps); $B \to A$ (1 step).

4.2 System β: Two Fixed Points

$R_O = \{A, B, C, D, E\}$, with ER defined as:

$$A \mapsto A, \quad B \mapsto A, \quad C \mapsto B, \quad D \mapsto E, \quad E \mapsto E.$$

Two persistent structures: A and E. Basin of A: $\{B, C\}$. Basin of E: $\{D\}$. Endpoint depends on starting state—illustrating why the stochastic result is needed.

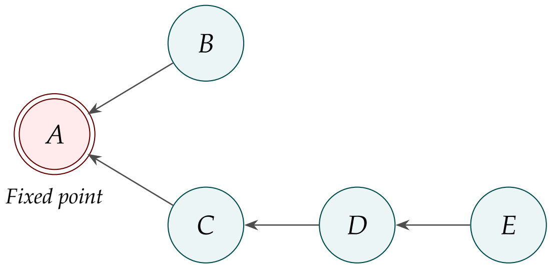

Figure 2: System α: all arrows flow toward A, the unique fixed point. The longest chain ($E \to D \to C \to A$) has length 3, so convergence takes at most 3 steps from any starting state.

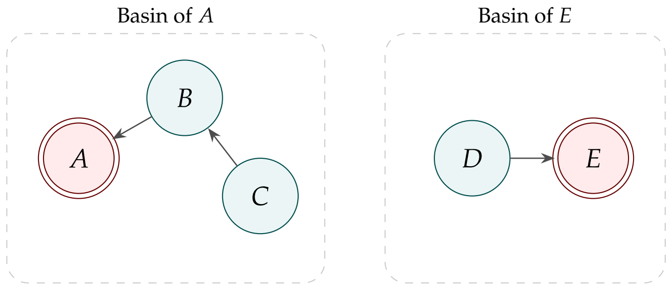

Figure 3: System β: two disconnected basins. Starting from C, the observer converges to A; from D, to E. Deterministic dynamics cannot cross from one basin to the other.

4.3 Numerical Walkthrough: System β with Noise

To illustrate the stochastic result concretely, consider System β with noise floor $\varepsilon = 0.05$. The transition matrix K is:

$$K = \begin{pmatrix} 0.80 & 0.85 & 0.05 & 0.05 & 0.05 \\ 0.05 & 0.05 & 0.80 & 0.05 & 0.05 \\ 0.05 & 0.05 & 0.05 & 0.05 & 0.05 \\ 0.05 & 0.00 & 0.05 & 0.05 & 0.05 \\ 0.05 & 0.05 & 0.05 & 0.80 & 0.80 \end{pmatrix}. \tag{3}$$

Each column represents the transition probabilities from a given state. The dominant entries correspond to the deterministic map E_R (e.g., column C has 0.80 in row B, reflecting $C \mapsto B$), with $\varepsilon = 0.05$ *noise spread across other states.*

The Dobrushin coefficient $\delta(K) \approx 0.75$. After $N = 2$ steps, $\delta(K^2) \approx 0.56 < 1$, confirming strict contraction. The unique stationary distribution is approximately:

$$\mu^* \approx (0.38,\ 0.12,\ 0.05,\ 0.05,\ 0.40). \tag{4}$$

From any initial distribution μ_0, the distance $d_{\mathrm{TV}}(\mu_0 K^n, \mu^*) \leq (0.56)^{\lfloor n/2 \rfloor}$. After 20 steps, the bound gives $d_{\mathrm{TV}} < 0.003$—effective convergence.

5 Deterministic Observers: Finite-Time Convergence

Lemma 5.1 (No Periodic Orbits). *Under* (C1), *E_R has no periodic orbits of period $k > 1$.*

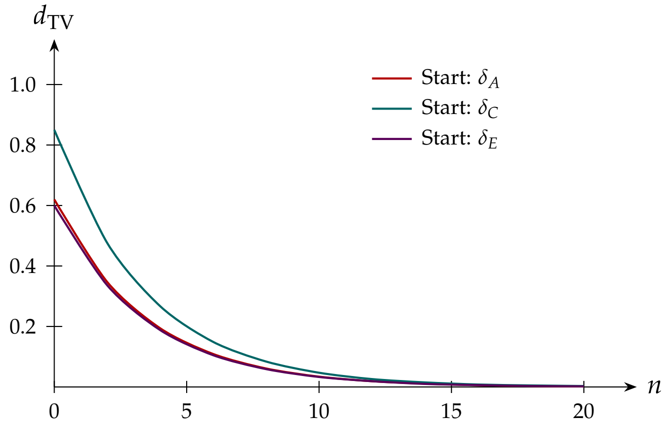

Figure 4: Convergence of $d_{\mathrm{TV}}(\mu_0 K^n, \mu^*)$ for three initial distributions on System β with noise $\varepsilon = 0.05$. All curves converge to zero, confirming that the initial state is forgotten.

Proof. Suppose $ER^k(x) = x$ for $k > 1$. Each orbit state is non-fixed. By (C1), chaining gives $H(ER^k(x)) < H(x)$. But $ER^k(x) = x$: contradiction. □

If you could return to a state you had already visited, entropy would have to both decrease (from C1) and stay the same (because you returned to the same state). This is impossible, so cycles are ruled out.

Cycle exclusion applies to the deterministic model only. In the stochastic model, noise permits backward transitions—enabling cross-basin mixing.

The Ruliad's raw computational dynamics contain limit cycles, gliders, and periodic attractors (Wolfram, 2002). Lemma 5.1 does not contradict this: E_R describes lossy compression, not raw computation. DPI-constrained compression cannot sustain cycles. The exclusion is a property of observation, not of the Ruliad itself.

Theorem 5.2 (Deterministic Convergence)**.** *Under* (C1)*, for every* $x \in R_O$*,* $\exists\, k \leq |R_O|$ *such that* $ER^k(x) \in \mathrm{Fix}(ER)$.

Proof. The orbit visits at most $|R_O|$ distinct states. No non-fixed state is revisited (Lemma 5.1). The orbit reaches a fixed point within $|R_O|$ steps. □

Interpretation. Every finite observer reaches equilibrium in bounded time. The bound depends only on $|R_O|$, not on environmental complexity. This is elementary—but its application to observer dynamics is the substantive claim: the act of observation itself terminates.

Analogy: a ball rolling downhill on a landscape with finitely many distinct altitudes must reach a valley within a bounded number of steps, because it loses altitude at each step and cannot revisit any altitude. The landscape is the observer's state space; the altitude is entropy.

Remark 5.3 (Why Not Banach). The Dobrushin coefficient of a deterministic transition matrix is generically 1. For a deterministic map ER, the transition matrix T has $T_{y,x} = 1$ if $ER(x) = y$ and 0 otherwise. For two states x_1, x_2 with $ER(x_1) \neq ER(x_2)$, the columns have their 1s in different rows, so $\min(T_{y,x_1}, T_{y,x_2}) = 0$ for all y, giving $\delta(T) = 1$. The finite-time argument (Lemma 5.1 + finiteness) is correct and stronger. Dobrushin is reserved for the stochastic setting, where non-zero off-diagonal entries guarantee $\delta < 1$.

The Dobrushin contraction coefficient is a tool designed for systems with randomness—where every state has some probability of reaching every other. A deterministic system, by definition, has no randomness, so the tool does not apply. We use a different, simpler, and actually stronger argument for the deterministic case.

Definition 5.4 (Pushforward)**.** $ER_*(\mu)(y) = \sum_{x:\, ER(x)=y} \mu(x)$.

Corollary 5.5 (Distribution Convergence)**.** *Under* (C1), $ER_*^n(\mu_0)$ *converges in* $\leq |R_O|$ *steps to* μ^* *supported on* $\mathrm{Fix}(ER)$.

Proof. By Theorem 5.2 and linearity of the pushforward. □

On System α: $\mu_0 = (0.2, 0.2, 0.2, 0.2, 0.2) \to \mu_3 = \delta_A$ in 3 steps.
On System β: $\mu_0 \to (0.6, 0, 0, 0, 0.4)$, stable on $\{A, E\}$. The limit depends on the initial distribution—uniqueness requires the stochastic result.

Corollary 5.6 (Ratchet Effect)**.** *For any persistent observer class,* $\mathrm{PS}(t_1) \subseteq \mathrm{PS}(t_2)$ *for* $t_1 \leq t_2$.

Proof. Fixed points are invariant under ER. Once discovered, they persist. □

Interpretation. Individual observers die, forget, degrade. But the class-level inventory only grows, because fixed points are features of the landscape. Mathematical truths survive dark ages. Under stochastic dynamics, fixed points are no longer perfectly absorbing—backward transitions create microscopic flux. The ratchet survives approximately: persistent structures carry overwhelming probability mass in μ^*, and the probability of losing a deeply stable structure is bounded by the vanishingly small backward transition rate. The ratchet holds exactly in the deterministic idealisation and approximately in the stochastic model.

6 Stochastic Observers: Exponential Convergence

6.1 Why the Stochastic Setting Is Needed

The deterministic result cannot answer two questions: (1) How fast does convergence occur? (2) Is the limit unique when $\mathrm{Fix}(ER)$ has multiple elements (System β)? The stochastic setting resolves both.

Definition 6.1 (Stochastic ER)**.** A primitive Markov matrix K on R_O: (a) K is stochastic (columns sum to 1, all entries ≥ 0); (b) K is irreducible and aperiodic.

Remark 6.2 (Computational Efficiency). Transition probabilities are biased toward lower-entropy states. Backward transitions are penalised by (B_O, Rel_O) but must be nonzero—justified by quantum uncertainty (Zurek, 2003). The primitivity index N is large for efficient observers: within-basin convergence is fast, cross-basin mixing is slow. The stochastic noise corresponds to the dissipative component (Γ) of the Helmholtz–Hodge decomposition in FEP, not the solenoidal component (Q)—see Section 9.3.

Definition 6.3 (Total Variation)**.** $d_{\mathrm{TV}}(\mu, \nu) = \frac{1}{2} \sum_x |\mu(x) - \nu(x)|$.

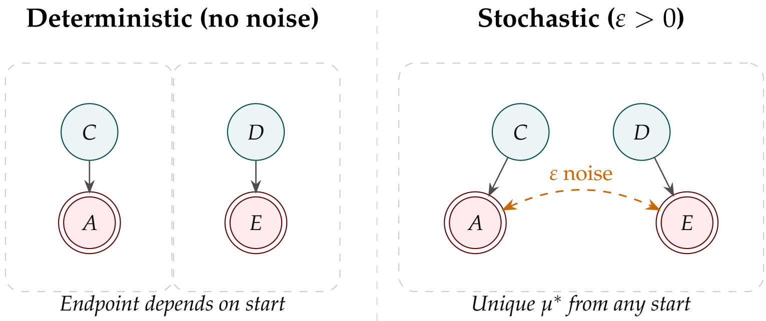

Figure 5: Convergence vs. uniqueness. *Left:* disconnected basins under deterministic dynamics—convergence occurs within each basin but the endpoint depends on starting state. *Right:* noise connects the basins, yielding a unique equilibrium from any initial condition.

Lemma 6.4 (Completeness). *$(\Delta(R_O), d_{\mathrm{TV}})$ is complete ($|R_O| < \infty \Rightarrow$ compact $\Rightarrow$ complete).*

The space of probability distributions on a finite set, equipped with total variation distance, is a complete metric space. This is the technical prerequisite for applying Banach's fixed-point theorem.

Definition 6.5 (Dobrushin Coefficient).

$$\delta(K) \;=\; 1 \;-\; \min_{x_1,x_2} \sum_y \min\big(K_{y,x_1},\; K_{y,x_2}\big). \tag{5}$$

The Dobrushin coefficient measures how much two different starting states can be distinguished after one step of the Markov chain. If $\delta = 0$, every pair of starting states leads to the same distribution (perfect mixing). If $\delta = 1$, some pair remains perfectly distinguishable (no mixing). For any primitive matrix, some power K^N achieves $\delta(K^N) < 1$.

Lemma 6.6 (Strict Contraction). *Under Definition 6.1, $\exists\, N \geq 1$ such that $\delta(K^N) < 1$.*

Proof. K primitive $\Rightarrow \exists\, N$ with $\varepsilon_N := \min_{x,y}(K^N)_{y,x} > 0$. Column overlap $\geq |R_O| \cdot \varepsilon_N$. Therefore $\delta(K^N) \leq 1 - |R_O|\, \varepsilon_N < 1$. □

Once K has been applied N times, every starting state has at least ε_N probability of reaching every other state. This forces a minimum overlap between any two resulting distributions, making the Dobrushin coefficient strictly less than 1.

Theorem 6.7 (Stochastic Convergence). *Under Definition 6.1, $\exists!\, \mu^*$ with $\mu^* K = \mu^*$, and:*

$$d_{\mathrm{TV}}(\mu_0 K^n,\; \mu^*) \;\leq\; \delta(K^N)^{\lfloor n/N \rfloor} \cdot d_{\mathrm{TV}}(\mu_0,\; \mu^*). \tag{6}$$

Proof. K_*^N is a strict contraction on the complete space $(\Delta(R_O), d_{\mathrm{TV}})$. Banach's Fixed-Point Theorem (Banach, 1922) gives unique μ^* of K_*^N. Shared fixed-point argument: $\mu^* K^N = \mu^* \Rightarrow \mu^* K = \mu^*$. □

Banach's theorem says: if a function on a complete metric space always brings points closer together (by a fixed ratio), then there is exactly one point that the function leaves unchanged, and repeated *application from any starting point converges to it. Here the "function" is the Markov transition applied N times, the "complete metric space" is the space of probability distributions, and the "fixed ratio" is $\delta(K^N) < 1$.*

Remark 6.8 (Convergence Timescales). For deeply stable structures, cross-basin mixing time may exceed cosmological timescales. The theorem guarantees existence of μ^* but the equilibrium may not be reachable within any finite horizon. Paradigm shifts are rare, violent, and slow; learning within an active paradigm is fast and smooth.

Remark 6.9 (Complementarity of the Two Results). The deterministic result governs fast dynamics (within-basin). The stochastic result governs slow dynamics (cross-basin). Together: fast dynamics converge to a neighbourhood; slow dynamics equilibrate to a point.

6.2 Non-Homogeneous Convergence

Lemma 6.10 (Uniform Bound)**.** *Under* (C2′)*,* $\delta(K_t^{N_{\max}}) \leq 1 - n_{\max}\,\varepsilon_{\min} =: \delta_{\max} < 1$ *for all* t*.*

Theorem 6.11 (Non-Homogeneous Convergence)**.** *Under* (C1), (C2′)*:*

$$d_{\mathrm{TV}}(\mu \cdot T_n,\ \nu \cdot T_n) \ \to\ 0 \qquad \textit{for all } \mu,\ \nu, \tag{7}$$

where $T_n = K_1 K_2 \cdots K_n$ *is the product of (possibly time-varying) transition matrices.*

Proof. Weak ergodicity (Seneta, 1981, Thm. 4.13) on $N_{\max}$-step blocks. $\sum(1 - \delta_{\max})$ diverges. □

Even if the environment changes over time (different transition matrices at each step), initial conditions are still forgotten, provided the noise floor persists. All observers track the same time-dependent distribution, regardless of where they started.

Remark 6.12 (What Theorem 6.11 Proves). Forgetting of initial conditions: all observers track the same time-dependent distribution. In a non-stationary environment, the target distribution may drift. Persistent structures under non-stationarity are features of the time-averaged dynamics.

7 Observer Completeness

Proposition 7.1 (Observer Completeness)**.** *For any finite collection* $\{O_i\}$ *with* $B_{O_i} < \infty$*,* $\exists\, O_M$ *with* $R_{O_M} \supseteq \bigcup_i R_{O_i}$*.*

Proof. The sampling functor S_M is a finite lookup table—a valid computational rule in $\mathbf{R}$ (Definition 2.1). □

Given any finite set of observers, there exists a "meta-observer" that can see everything any of them can see. This meta-observer exists as a rule in the Ruliad—it is a valid computation—but it may not be physically instantiable.

Remark 7.2 (Physical Realisability). O_M exists as a rule in the Ruliad but may not be physically instantiable: B_{O_M} may exceed the computational capacity of the physical universe. Mathematical existence (in the Ruliad) $\neq$ physical instantiation (in a particular universe).

8 Categorical Interpretation: Structural Unity

Having established that observers converge dynamically to persistent structures (Theorem 5.2–6.11), we note a structural consequence: the persistent structures to which observers converge are all connected to TI, the terminal object in the Ruliad's categorical formalism, via unique morphisms.

This connection is *structural*, not dynamical. It follows from the definition of TI as terminal in **R** (Definition 2.4), not from the convergence theorems. The convergence theorems tell us that observers *reach* persistent structures; the terminal object's defining property tells us that persistent structures *point toward* TI. These are logically independent facts with independent sources.

Observation 8.1 (Structural Connection to TI). Let O be an observer satisfying *(C1)*. Then:

(i) $\mathrm{Fix}(ER_O)$ is non-empty (by Theorem 5.2), and every $x \in \mathrm{Fix}(ER_O)$ is an object in **R**.

(ii) By Definition 2.4, there exists a unique morphism $f_x\colon x \to \mathrm{TI}$ for each such x.

(iii) The collection of morphisms $\{f_x\}_{x\in\mathrm{Fix}(ER_O)}$ is the unique family connecting O's persistent structures to TI.

Non-emptiness of $\mathrm{Fix}(ER_O)$ is the sole contribution of the convergence theorems to this observation: Theorem 5.2 guarantees that every orbit of ER terminates at a fixed point. That fixed points are objects in **R** is definitional. Statements (ii) and (iii) follow from TI's terminality and the uniqueness clause in the definition of terminal object.

Remark 8.2 (What This Does and Does Not Prove). Observation 8.1 does *not* establish that the relationship between different observers' persistent structures is governed by the convergence dynamics. The morphisms $f_x\colon x \to \mathrm{TI}$ exist for any object in **R**—persistent or not, observed or not. What the convergence theorems contribute is the guarantee that observers *arrive at* persistent structures: without Theorem 5.2, there is no reason to believe that the orbit of ER terminates, and therefore no reason to apply the structural observation to observer dynamics specifically.

The substantive mathematical content of this paper resides entirely in the convergence theorems (Sections 5–6). The structural observation provides the interpretive frame connecting those results to the God Conjecture's broader claims about structural unity.

8.1 Connectivity via Meta-Observers

Remark 8.3 (Structural Bridging). Proposition 7.1 guarantees that for any two observers O_1, O_2, there exists a meta-observer O_M with $R_{O_M} \supseteq R_{O_1} \cup R_{O_2}$. Combined with Observation 8.1, this means: for any two observers' persistent structures $x_1 \in \mathrm{Fix}(ER_1)$ and $x_2 \in \mathrm{Fix}(ER_2)$, the morphisms f_{x_1} and f_{x_2} to TI factor through the same ambient category **R**. Different observers' endpoints are structurally related—not because of a shared dynamical mechanism, but because TI's terminality provides a common reference point.

This is a weaker claim than the God Conjecture's original formulation, which posited a dynamical mechanism connecting all observer equilibria. The dynamical version requires a richer categorical framework—in particular, a category of observer equilibria equipped with morphisms that respect the entropy reduction dynamics (ER-naturality). That construction, including the relationship between TI as defined (terminal in **R**) and TI as constructed (colimit of the directed system of persistent structures), is the subject of a companion paper (Senchal, 2026b).

8.2 The Main Result

Theorem 8.4 (Universal Convergence)**.** *Let* $\{O_i\}$ *be observers with* $B_{O_i} < \infty$*. Under* (C1), (C2′)*:*

(i) ***Local convergence.*** *Each* O_i *converges to* $\mu^*_{O_i}$ *on* $\mathrm{Fix}(ER_i)$*.*

(ii) ***Structural connection.*** *For each persistent structure* $x \in \mathrm{Fix}(ER_i)$*, there exists a unique morphism* $f_x \colon x \to \mathrm{TI}$ *in* **R**.

(iii) ***Ratchet.*** $\mathrm{PS}(t)$ *is non-decreasing for persistent classes.*

Proof. (i): Theorem 5.2, 6.7, 6.11. (ii): Definition 2.4 (terminality of TI in **R**). (iii): Corollary 5.6. □

Interpretation. Part (i) is the mathematical substance—it is proved, non-trivially, from the dynamics. Part (ii) is a structural observation that follows from what TI is. Part (iii) is proved from the dynamics. The theorem as stated is honest about this separation: it does not claim that the structural connection in (ii) is a consequence of the convergence in (i). They are independent facts that, together, support the God Conjecture's claim that all observers converge toward structurally related persistent structures.

The theorem says: every observer converges (from the dynamics), and all the things they converge to point to the same place (from the category theory). The convergence is real and proven; the "pointing" is structural and follows from what TI *is.*

Remark 8.5 (The Status of TI). TI functions as a mathematical boundary condition—a formal point at infinity. Whether it has ontological status beyond this structural role is a question the formalism makes precise but does not answer. The convergence theorems do not depend on the answer. The companion paper's colimit construction (Senchal, 2026b) addresses this by constructing TI from below—as the totality of what observers converge on—rather than imposing it from above. Whether the two objects coincide is an open question with interpretive consequences (see Section 9.6).

9 Discussion

9.1 The Two-Source Insight

Convergence is architecture-independent (depends only on finiteness). Structural connection is substrate-independent (depends only on TI's terminality in **R**, which is definitional). Together: any finite observer in any connected universe converges on persistent structures related to the same categorical limit. These are logically independent facts with independent physical sources.

9.2 Relationship to the Free Energy Principle

Observer Theory captures only the gradient/dissipative component of NESS dynamics (Da Costa et al., 2021). The FEP's attracting-set result is arguably more physically realistic for biological observers than Theorem 6.7's unique-point result, which should be understood as an infinite-time idealisation. The Markov dynamics describe epistemic evolution, not underlying physics; the quantum structure of the Ruliad is captured at the multiway/branchial level (Gorard and Arsiwalla, 2021).

The formal equivalence between Observer Theory and the FEP (Senchal, 2025a, §6) provides independent confirmation at the level of convergence results. The Helmholtz–Hodge decomposition from FEP further decomposes dynamics into a gradient (dissipative) component and a solenoidal (conservative/exploratory) component. The convergence theorems in this paper capture the gradient component. The solenoidal component—deterministic, entropy-preserving circulation driving active inference—is not captured by the current formalism and represents the most important open question. This open question does not change this result (Section 9.3).

9.3 Helmholtz–Hodge Decomposition and the Solenoidal Gap

In Friston's formalism (Da Costa et al., 2021; Friston, 2019), NESS dynamics decompose as:

$$f(x) \;=\; -(\Gamma + Q)\,\nabla I(x), \tag{8}$$

where:

- **Gradient/dissipative** ($-\Gamma\nabla I$): driven by noise Γ, performs gradient descent on surprisal. Entropy-reducing, convergent.
- **Solenoidal/conservative** ($Q\nabla I$): mediated by antisymmetric Q, circulates deterministically on iso-surprisal contours. Entropy-preserving. Drives active inference.

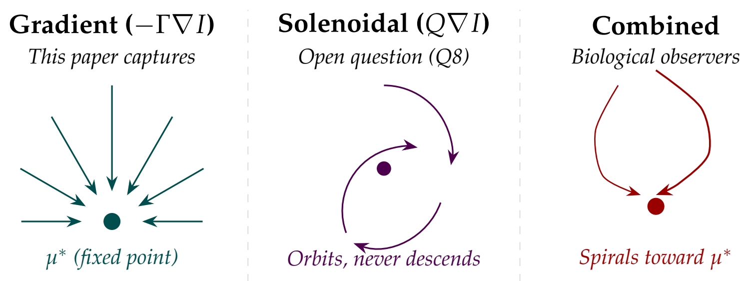

Figure 6: Helmholtz–Hodge decomposition. *Left:* the gradient component—a vector field pointing downhill toward μ^* (captured by the convergence theorems). *Centre:* the solenoidal component—circulation that orbits without descending (not captured by the current formalism). *Right:* the combined dynamics—a spiral that both descends and orbits, characteristic of biological observers.

The convergence theorems capture the gradient component. The stochastic noise in K corresponds to Γ—dissipative mixing. The solenoidal component Q has no analogue in the current formalism: Lemma 5.1 excludes deterministic cycles, and noise corresponds to Γ-driven diffusion, not Q-mediated circulation.

Where the solenoidal component would live. In the ∞-groupoid structure (Gorard and Arsiwalla, 2021), vertical morphisms (computational reduction) correspond to the gradient; horizontal morphisms (branchial exploration at constant depth) correspond to the solenoidal component. These are categorically orthogonal, which is why convergence is unaffected by the solenoidal component's absence.

The solenoidal gap is the paper's most important limitation. The formalism captures convergent sensemaking but not directed exploration. Current LLMs share this limitation. Biological observers possess both. Formalising the solenoidal component is the most pressing open question (Q8).

9.4 The Ratchet, Civilisational Collapse, and Dark Ages

Persistent structures are features of the dynamics, not of any observer. The Pythagorean theorem was proved independently in Greece, China, and India—not via cultural transmission, but because it is a persistent structure of the accessible Ruliad.

9.5 Convergence and Alignment

The solenoidal gap has direct implications for AI alignment (Senchal, 2026a). The precise architectural absence is Q-mediated directed exploration (i.e. it is not "noise" or "randomness"). This formalism makes the gap precise.

9.6 On the Theological Interpretation

The gap between "persistent structures map to TI" and "TI is what mystics describe" requires the C_F/C_R decomposition distinguishing convergence from shared cognition (Atran, 2002; Boyer, 2001) versus convergence from shared reality structure. This is the weakest point of the current argument and the subject of forthcoming work.

The stronger claim—that TI can be *constructed* from observer dynamics rather than presupposed—is addressed in a companion paper (Senchal, 2026b), which constructs TI as a filtered colimit of the directed system of persistent structures and examines whether this constructed object coincides with the definitional TI of Definition 2.4. If they coincide, reality is fully discoverable; if not, there exists structure beyond the reach of any finite observer. Both possibilities have precedents in the traditions under discussion.

9.7 Open Questions

(Q1) Computing Dobrushin coefficients for specific architectures.

(Q2) Formalising inter-observer networking effects.

(Q3) Applying discrete Hodge theory to observer state spaces.

(Q4) Lyapunov function on the $(\infty, 1)$-topos.

(Q5) Experimental validation on multiway systems.

(Q6) C_F/C_R cognitive architecture decomposition.

(Q7) Thermodynamic cost of Rel_O evaluation in the neighbourhood the Observer computes over (Landauer's principle).

(Q8) Formalisation of the solenoidal component: entropy-preserving horizontal morphisms and their relationship to Q in Friston's HHD. *This is the most important open question.*

(Q9) Constructing TI as the filtered colimit of observer persistent structures ($\mathrm{TI}^{\mathrm{col}} = \mathrm{colim\ PS}$), equipped with ER-natural morphisms, and determining whether $\mathrm{TI}^{\mathrm{col}} \cong \mathrm{TI}^{\mathrm{def}}$ (Definition 2.4). See Senchal (2026b).

10 Conclusion

The God Conjecture's central claim—that all finite observers converge toward structurally connected persistent structures—is now a theorem. The convergence is dynamical (Theorem 5.2–6.11), proved via finite-time termination for deterministic observers and Dobrushin contraction for stochastic observers (extending to non-stationary environments under weak ergodicity). The structural connection is categorical (Theorem 8.4): all persistent structures admit unique morphisms to TI by its terminality. The convergence theorems provide the dynamical substance; TI's terminality provides the structural interpretation. The full categorical treatment—constructing TI from observer dynamics and equipping the category of observer equilibria with ER-natural morphisms—is deferred to a companion paper (Senchal, 2026b).

The result holds under one physical assumption: the universe has stable statistical properties. Everything else follows from definitions or is proved from them.

The most important open question is the formalisation of the solenoidal component—the deterministic, entropy-preserving exploration that distinguishes biological observers from purely dissipative systems. The convergence theorems capture the gradient dynamics: the descent toward persistent structures. The solenoidal dynamics—the exploration that determines the richness of the descent—are not yet formalised.

The river reaches the sea. The question is what it discovers along the way.

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* sam@maddoxcp.com